

HeatGeo: Corpus-Level Heat-Kernel Geometry Distillation for Compact Text Embedding Models

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Abstract

Large text embedding models are central to semantic search, clustering, and retrieval-augmented generation (RAG), where quality depends on neighborhood structure over an indexed corpus. Existing embedding distillation methods transfer local teacher signals, from embeddings and token states to batch-level similarities. However, in retrieval and RAG, an embedding model is judged after it becomes an index: small distortions in corpus-level neighborhoods can change which evidence is surfaced, even when individual student vectors remain close to the teacher. This exposes a mismatch between local distillation losses and the neighborhood dynamics retrieval depends on. The key object to preserve is therefore not a set of vectors, but the teacher’s retrieval operator over the corpus. We propose **HeatGeo**, a corpus-level geometry distillation framework that models this behavior as a semantic random walk on a teacher-induced neighborhood graph. HeatGeo caches teacher embeddings, builds a mutual k -nearest-neighbor graph, and derives multi-scale heat-kernel diffusion distributions describing how semantic mass spreads from each example to nearby regions and higher-order communities. The student matches these distributions over sampled candidate neighborhoods, while a spectral manifold objective anchors global topology using graph Laplacian eigenvectors. We evaluate on STS, text classification, and pair classification to test whether corpus-level dynamics preserve retrieval geometry better than pointwise, token-level, and mini-batch relational baselines.

1 Introduction

Text embedding models have become a basic infrastructure layer for semantic search, clustering, reranking, and retrieval-augmented generation. In these systems, a model is not judged only by whether it embeds one sentence close to a teacher vector; it is judged by whether the resulting corpus index organizes examples into useful neighborhoods. Strong recent embedding models, including LLM-based encoders, improve this organization but are costly to run at scale. Encoding large private corpora, refreshing indexes as documents change, or serving retrieval on constrained hardware therefore motivates compact student encoders that inherit the semantic structure of a stronger teacher.

Knowledge distillation offers a practical route to this compression (Hinton, Vinyals, and Dean 2015). In language representation learning, distillation has been applied through output matching, hidden-state alignment,

self-attention transfer, and compact pretrained architectures (Sanh et al. 2019; Jiao et al. 2020; Wang et al. 2020; Xu et al. 2024). For embedding models, common objectives include pointwise alignment between student and teacher embeddings, contrastive learning with teacher-derived positives, and relational losses that match pairwise similarities within a mini-batch. These objectives are attractive because they are simple and efficient. They also fit naturally into stochastic training.

However, embedding models are deployed in a different regime from the one these losses directly supervise. Retrieval and clustering depend on the neighborhood structure induced over an entire corpus. A pointwise loss asks the student to reproduce teacher coordinates, even when a compact student may not have enough capacity to match the teacher’s absolute geometry. A mini-batch relational loss improves on this by comparing local similarity patterns, but it sees only the examples that happen to co-occur in a batch. Many relations that matter for retrieval are neither pointwise nor confined to a single batch: samples may be connected through multi-hop semantic paths, and useful clusters often emerge from the accumulated geometry of the full corpus.

This mismatch motivates the central question of this work: can we distill a large embedding teacher by matching its corpus-level semantic diffusion process, rather than only matching individual vectors or mini-batch relations? The intuition is simple. If the teacher’s embedding space defines a semantic manifold over the training corpus, then random walks on a teacher-induced neighborhood graph reveal how semantic mass flows from each example to nearby examples, broader local regions, and higher-order communities. Matching this diffusion process gives the student direct supervision on the retrieval geometry it must later reproduce.

We propose **HeatGeo**, a corpus-level heat-kernel geometry distillation method for compact text embedding models. HeatGeo first caches teacher embeddings for the training corpus and constructs a mutual k -nearest-neighbor graph using teacher cosine similarity. From this graph, it defines a transition matrix and precomputes multi-scale diffusion targets corresponding to one-step, two-step, and longer-range semantic random walks. During training, each anchor example is paired with a candidate set containing high-diffusion neighbors, teacher hard negatives, and random negatives. The student then learns a neighborhood distribution over the

same candidates and minimizes a weighted KL divergence to the teacher diffusion targets.

HeatGeo adds two stabilizing signals. First, a spectral manifold objective projects student embeddings into low-dimensional spectral coordinates computed from the teacher graph Laplacian, encouraging the student to preserve global semantic topology rather than only local neighbors. Second, a lightweight pointwise anchor aligns projected student embeddings with cached teacher embeddings, acting as a boundary condition rather than the main training signal. A simple curriculum introduces diffusion scales gradually: the student first learns local neighborhoods, then semantic regions, and finally longer-range diffusion with spectral anchoring.

Our contributions are:

- We formulate text embedding distillation as a corpus-level geometry transfer problem, highlighting the gap between common pointwise or mini-batch losses and corpus-level retrieval behavior.
- We introduce HeatGeo, which distills multi-scale heat diffusion distributions from a teacher-induced mutual k NN graph into a compact student encoder.
- We propose a spectral manifold anchoring objective that transfers global topology from the teacher graph without requiring teacher hidden states or modifying the student architecture.
- We provide an evaluation protocol for HeatGeo across semantic textual similarity, text classification, and pair classification benchmarks, with ablations for diffusion scale, spectral anchoring, candidate sampling, and curriculum.

2 Related Work

Text Embedding Models and Evaluation. Text embedding models encode variable-length text into fixed-dimensional vectors for retrieval, clustering, semantic textual similarity, and transfer learning. Sentence-BERT made transformer encoders practical for sentence-level similarity search by using siamese and triplet-network training (Reimers and Gurevych 2019), while SimCSE showed that contrastive learning with dropout noise or entailment supervision can produce strong sentence embeddings (Gao, Yao, and Chen 2021). Evaluation suites such as BEIR and MTEB further shifted the field from single-task similarity scores toward broad retrieval, classification, clustering, and semantic textual similarity evaluation (Thakur et al. 2021; Muennighoff et al. 2023). Continued maintenance of MTEB reflects the need for reproducible, multi-task embedding evaluation as models and datasets evolve (Chung et al. 2025).

Recent embedding systems increasingly rely on larger foundation models, richer training mixtures, and specialized retrieval objectives. BGE-M3 combines multilingual, multi-functionality, and multi-granularity training with self-knowledge distillation across dense, sparse, and multi-vector retrieval modes (Chen et al. 2024). Qwen3 Embedding scales embedding and reranking models from Qwen3 foundation models (Zhang et al. 2025), and Dewey targets long-

context embedding for RAG-style retrieval (Zhang, Zou, and Zhou 2025). This line of work provides stronger teachers and broader evaluation targets, but it does not by itself specify how a compact student should preserve the teacher’s corpus-level neighborhood organization. HeatGeo treats these models as black-box embedding teachers: it requires only their final embeddings, then distills the semantic graph induced over the training corpus.

Distillation for Language and Embedding Models.

Knowledge distillation transfers behavior from a large teacher to a smaller student (Hinton, Vinyals, and Dean 2015). In pretrained language models, DistilBERT, TinyBERT, and MiniLM compress transformer encoders through combinations of logits, hidden states, attention distributions, and self-attention relations (Sanh et al. 2019; Jiao et al. 2020; Wang et al. 2020). Recent LLM distillation extends this toolkit with on-policy training, contrastive objectives, sparse-logit acceleration, and score-matching formulations (Agarwal et al. 2024; Ko et al. 2025; Anshumann et al. 2025; Kim et al. 2025; Xu et al. 2024). Cross-tokenizer methods such as DSKD and CDM address mismatches between teacher and student vocabularies or tokenization schemes (Zhang et al. 2024b; Chen et al. 2025).

Embedding-specific distillation adapts these ideas to retrieval-oriented representations. Gecko distills LLM-generated and LLM-relabelled retrieval data into compact text embedding models (Lee et al. 2024). Jasper and Stella use multi-stage embedding distillation and Matryoshka Representation Learning to improve both quality and dimensional flexibility (Zhang et al. 2024a). LEAF aligns compact embedding students to teacher representations and supports asymmetric retrieval in which queries and documents may be encoded by different models (Vujanic and Ruecksties 2025). Jina Embeddings v5 combines task-targeted distillation with contrastive training to produce compact multilingual embeddings robust to truncation and quantization (Akram et al. 2026). These methods establish distillation as a central recipe for embedding model development. Their supervision, however, is still primarily defined by individual teacher vectors, generated/relabelled pairs, task labels, dimensional heads, or local teacher scores. HeatGeo changes the distillation object from per-example or per-pair behavior to multi-scale diffusion over a teacher-induced corpus graph.

Local and Relational Supervision. Pointwise embedding distillation directly matches a projected student embedding to the corresponding teacher embedding. This is simple and architecture-agnostic, but it asks a compact student to reproduce the teacher’s absolute coordinates even when the student’s capacity and embedding dimension differ. Contrastive sentence-embedding distillation methods such as DistilCSE improve the training signal by transferring contrastive behavior and pairwise preferences (Gao et al. 2021; Xu et al. 2023). Relational knowledge distillation further argues that students should preserve relations among examples rather than only individual outputs (Park et al. 2019). In embedding training, this idea commonly becomes mini-batch similarity matching: teacher and student pairwise similarity matrices

are compared over the examples that happen to co-occur in a stochastic batch.

These local objectives better reflect retrieval than pure coordinate regression, but their supervision is bounded by the sampled batch or pair set. They do not explicitly model multi-hop semantic paths, community structure, or the way probability mass should spread from an anchor to nearby regions of the full corpus. Token-level and layer-level alignment methods, including CKA-style representation matching, optimal-transport alignment, EMO-style objectives, and teacher-anchored layer alignment, provide useful intermediate supervision when teacher internals or student layers are available (Kornblith et al. 2019; Alqahtani et al. 2021; Dao et al. 2026). Yet they still optimize local representational agreement rather than a corpus-level retrieval geometry. HeatGeo keeps the useful architecture-agnostic property of final-embedding distillation, but replaces mini-batch relational snapshots with cached diffusion targets sampled from the full teacher graph.

Graph, Manifold, and Diffusion Perspectives. Graph-based manifold learning studies how neighborhood graphs reveal lower-dimensional structure in high-dimensional data. Laplacian Eigenmaps use graph Laplacian eigenvectors to preserve local neighborhoods in low-dimensional representations (Belkin and Niyogi 2003), and Diffusion Maps use Markov diffusion on a graph to expose multi-scale connectivity and geometry (Coifman and Lafon 2006). Knowledge distillation on graphs also studies how structural information can be transferred between graph models or from graph teachers to simpler students (Tian et al. 2025; Zhang et al. 2022). These works show that graph structure and diffusion can encode information unavailable from isolated examples.

HeatGeo draws on this graph-manifold view but applies it to a different distillation setting. The graph is not an observed input graph and the spectral coordinates are not used as a standalone dimensionality-reduction output. Instead, the graph is induced by a strong text embedding teacher over the training corpus, diffusion distributions become soft neighborhood targets for the student, and Laplacian eigenvectors act as a global topology anchor. This directly addresses the gap left by pointwise, token-level, layer-level, and mini-batch relational distillation: compact text encoders are trained to preserve the teacher’s corpus-level semantic diffusion process, the object most closely aligned with retrieval and clustering behavior.

3 Methodology

3.1 Problem Formulation

Let $\mathcal{D} = \{x_i\}_{i=1}^N$ denote the unlabeled distillation corpus. A teacher encoder T maps each input to a cached teacher embedding

$$t_i = T(x_i) \in \mathbb{R}^{d_T}, \quad (1)$$

and a compact student encoder S_θ maps the same input to

$$s_i = S_\theta(x_i) \in \mathbb{R}^{d_S}. \quad (2)$$

The objective is to train S_θ so that the student preserves the semantic organization induced by the teacher over the corpus, including nearest-neighbor rankings, local semantic regions, and higher-order community structure.

Standard pointwise embedding distillation minimizes a distance between each projected student embedding and its teacher embedding:

$$\mathcal{L}_{\text{point}} = \frac{1}{N} \sum_{i=1}^N (1 - \cos(W s_i, t_i)), \quad (3)$$

where W is an optional projection. Relational distillation instead matches student and teacher similarity matrices inside a mini-batch B :

$$\mathcal{L}_{\text{rel}} = \|G_B^S - G_B^T\|_F^2, \quad (4)$$

where G_B^S and G_B^T are cosine-similarity matrices over examples in B . These objectives are useful local surrogates, but they do not directly supervise the corpus-level retrieval operator induced by the teacher.

3.2 Motivation

HeatGeo starts from the observation that retrieval quality depends on how probability mass is organized over an indexed corpus, not only on the absolute coordinate of each embedding. A compact student may be unable to reproduce the teacher’s full embedding map, yet it can still be a strong retrieval model if it preserves the teacher’s neighborhood distributions and global semantic topology.

We therefore treat the teacher embedding space as a semantic graph and distill the random-walk behavior of this graph. For an anchor x_i , a one-hop walk captures direct semantic neighbors; multi-hop walks reveal broader regions connected through intermediate examples. This makes the distillation target a soft corpus-level object: how semantic mass should spread from an anchor to nearby candidates at different scales.

For a diffusion scale r , write the full teacher diffusion row as

$$q_{i,r}(j) = (P_r^T)_{ij}, \quad j \in \mathcal{D}. \quad (5)$$

Since matching a distribution over all N examples is expensive, HeatGeo samples a candidate set C_i and uses the conditional diffusion distribution on that set. Its captured mass is

$$\rho_{i,r}(C_i) = \sum_{j \in C_i} q_{i,r}(j). \quad (6)$$

The method is designed so that C_i contains high-diffusion neighbors, hard negatives, and random negatives, making the restricted target informative while keeping training tractable.

The rest of the section defines the two geometric transfer losses and the stabilizing anchor objective. Formal proofs for the theoretical statements are provided in Appendix A.

3.3 Multi-Scale Diffusion Loss

HeatGeo first encodes the full training corpus with the teacher and constructs a teacher semantic graph. For each example x_i , let

$$\mathcal{N}_k^T(i) = \text{TopK}_{j \neq i} \cos(t_i, t_j) \quad (7)$$

be its top- k teacher nearest neighbors. To reduce noisy one-directional edges and hubness, HeatGeo uses a mutual k NN graph:

$$j \in \mathcal{M}_k^T(i) \iff j \in \mathcal{N}_k^T(i) \text{ and } i \in \mathcal{N}_k^T(j). \quad (8)$$

The weighted adjacency matrix is

$$W_{ij}^T = \begin{cases} \exp(\cos(t_i, t_j)/\tau_g), & j \in \mathcal{M}_k^T(i), \\ 0, & \text{otherwise,} \end{cases} \quad (9)$$

where τ_g controls graph sharpness. Let D^T be the diagonal degree matrix. The row-normalized transition matrix is

$$P^T = (D^T)^{-1}W^T. \quad (10)$$

HeatGeo then defines multi-scale diffusion targets

$$P_r^T = (P^T)^r, \quad r \in \mathcal{R}, \quad (11)$$

where $\mathcal{R}$ is typically $\{1, 2, 4\}$. Small r preserves sharp local neighborhoods, while larger r captures semantic regions and multi-hop communities.

For each anchor x_i , HeatGeo builds a candidate set

$$C_i = C_i^+ \cup C_i^h \cup C_i^r, \quad (12)$$

where C_i^+ contains high-probability diffusion neighbors, C_i^h contains teacher hard negatives, and C_i^r contains random negatives. The teacher diffusion target at scale r is the diffusion mass restricted and renormalized over C_i :

$$p_{i,r}^T(j) = \frac{(P_r^T)_{ij}}{\sum_{j' \in C_i} (P_r^T)_{ij'}}, \quad j \in C_i. \quad (13)$$

Given an anchor embedding s_i and candidate embeddings $\{s_j : j \in C_i\}$, the student induces a distribution over the same candidate set:

$$p_i^S(j) = \frac{\exp(\cos(s_i, s_j)/\tau_s)}{\sum_{j' \in C_i} \exp(\cos(s_i, s_{j'})/\tau_s)}, \quad j \in C_i, \quad (14)$$

where τ_s is the student temperature. The diffusion loss is

$$\mathcal{L}_{\text{diff}} = \frac{1}{|B|} \sum_{i \in B} \sum_{r \in \mathcal{R}_e} \omega_r D_{\text{KL}}(p_{i,r}^T \| p_i^S), \quad (15)$$

where ω_r is the scale weight and $\mathcal{R}_e$ is the set of active scales at epoch e .

Lemma 1 (Diffusion KL preserves neighborhood mass). Fix an anchor i , a diffusion scale r , and candidate set C_i . If

$$D_{\text{KL}}(p_{i,r}^T \| p_i^S) \leq \delta, \quad (16)$$

then, for every subset $A \subseteq C_i$,

$$|p_i^S(A) - p_{i,r}^T(A)| \leq \sqrt{\frac{\delta}{2}}. \quad (17)$$

Implication for HeatGeo. The subset A can be any semantic slice of the candidate neighborhood: high-diffusion positives, teacher hard negatives, random negatives, a local cluster inside C_i , or a retrieval prefix induced by a downstream ranking rule. Thus $\mathcal{L}_{\text{diff}}$ preserves how much semantic probability mass the teacher assigns to regions of the

retrieval neighborhood, rather than merely aligning isolated vectors.

Lemma 2 (Candidate restriction under high captured mass). Let $q_{i,r}(j) = (P_r^T)_{ij}$ be the full-corpus teacher diffusion row, let $\rho = \rho_{i,r}(C_i) > 0$, and let $p_{i,r}^T$ be the restricted target on C_i . Define the zero extension of the student candidate distribution to the corpus by

$$\hat{p}_i^S(j) = \begin{cases} p_i^S(j), & j \in C_i, \\ 0, & j \notin C_i. \end{cases} \quad (18)$$

Then

$$\text{TV}_{\mathcal{D}}(q_{i,r}, \hat{p}_i^S) \leq (1 - \rho) + \rho \text{TV}_{C_i}(p_{i,r}^T, p_i^S). \quad (19)$$

Consequently, if $\rho_{i,r}(C_i) \geq 1 - \epsilon$ and $D_{\text{KL}}(p_{i,r}^T \| p_i^S) \leq \delta$, then

$$\text{TV}_{\mathcal{D}}(q_{i,r}, \hat{p}_i^S) \leq \epsilon + \sqrt{\frac{\delta}{2}}. \quad (20)$$

Implication for HeatGeo. The restricted target $p_{i,r}^T$ is the teacher diffusion row conditioned on the sampled candidates. Lemma 2 shows that the candidate approximation is faithful when C_i captures most diffusion mass: the omitted tail controls the unavoidable error, and the learned KL term controls the mismatch inside C_i .

3.4 Spectral Manifold Anchoring

Diffusion matching transfers local and multi-hop neighborhoods, but a student can still fit candidate distributions while distorting broader community structure. HeatGeo therefore computes spectral coordinates from the teacher graph. Let the symmetric normalized graph Laplacian be

$$L^T = I - (D^T)^{-1/2}W^T(D^T)^{-1/2}. \quad (21)$$

Let $U^T \in \mathbb{R}^{N \times m}$ contain the first m non-trivial eigenvectors of L^T , and let u_i^T be the row corresponding to x_i . A learnable projection maps student embeddings to this spectral space:

$$z_i^S = W_{\text{spec}} s_i. \quad (22)$$

The spectral loss is

$$\mathcal{L}_{\text{spec}} = \frac{1}{|B|} \sum_{i \in B} \|z_i^S - u_i^T\|_2^2. \quad (23)$$

Proposition 3 (Spectral anchoring preserves low-frequency topology). Fix a mini-batch B and a fixed choice of signs and basis for the teacher Laplacian eigenvectors in U^T . If

$$\mathcal{L}_{\text{spec}}(B) = \frac{1}{|B|} \sum_{i \in B} \|z_i^S - u_i^T\|_2^2 \leq \eta, \quad (24)$$

then the projected student coordinates preserve teacher spectral coordinates in mean square. Moreover, for every pair $i, j \in B$,

$$\begin{aligned} & \left| \|z_i^S - z_j^S\|_2 - \|u_i^T - u_j^T\|_2 \right| \\ & \leq \|z_i^S - u_i^T\|_2 + \|z_j^S - u_j^T\|_2, \end{aligned} \quad (25)$$

and therefore

$$\begin{aligned} & \frac{1}{|B|^2} \sum_{i,j \in B} \left| \|z_i^S - z_j^S\|_2 - \|u_i^T - u_j^T\|_2 \right| \\ & \leq 2\sqrt{\eta}. \end{aligned} \quad (26)$$

Implication for HeatGeo. The first non-trivial eigenvectors of the normalized graph Laplacian are low-frequency modes of the teacher semantic graph. They vary smoothly over strongly connected regions and separate broader communities, following the graph-manifold principles behind Laplacian Eigenmaps and Diffusion Maps (Belkin and Niyogi 2003; Coifman and Lafon 2006). The spectral loss therefore anchors the student to the teacher’s global topology.

3.5 Optimization Objectives

HeatGeo includes a lightweight pointwise anchor for stability:

$$\mathcal{L}_{\text{anchor}} = \frac{1}{|B|} \sum_{i \in B} (1 - \cos(W_a s_i, t_i)), \quad (27)$$

where W_a maps student embeddings to the teacher dimension. This term acts as a boundary condition on the embedding map, while the diffusion and spectral objectives carry the main geometric supervision.

The final objective is

$$\mathcal{L}_{\text{HeatGeo}} = \lambda_1 \mathcal{L}_{\text{diff}} + \lambda_2 \mathcal{L}_{\text{spec}} + \lambda_3 \mathcal{L}_{\text{anchor}}. \quad (28)$$

We use a diffusion-scale curriculum: the first epoch uses one-step diffusion, the second epoch adds two-step diffusion, and later epochs add the full scale set and spectral anchoring. This avoids forcing long-range, smoother structure before the student has learned reliable local neighborhoods.

Corollary 4 (Joint HeatGeo geometry transfer). Consider a mini-batch B and active diffusion scales $\mathcal{R}_e$. Assume all loss weights are positive and the candidate sets capture high teacher diffusion mass, i.e., $\rho_{i,r}(C_i) \geq 1 - \epsilon$ for all $i \in B$ and $r \in \mathcal{R}_e$. If the weighted HeatGeo objective is small on B , then the student simultaneously controls expected candidate-neighborhood mass error, expected full-corpus diffusion error up to the omitted tail ϵ , pairwise distances in the teacher spectral coordinate system, and the pointwise anchor mismatch.

Implication for HeatGeo. This result formalizes the role of the three losses. The diffusion loss transfers local and multi-hop retrieval behavior, candidate restriction makes the transfer computationally feasible when captured mass is high, the spectral loss anchors low-frequency manifold topology, and the anchor loss stabilizes the coordinate map. The guarantee is conditional and average-case; it does not claim exact preservation of every full-corpus distance.

4 Experiments

4.1 Experimental Setup

Benchmarks and metrics. Following the evaluation protocol used in TALAS (Dao et al. 2026), we evaluate HeatGeo on three embedding-reliant task families. For text classification, we report macro-F1 on BANKING77 (Casaneuva et al. 2020), Emotion (Saravia et al. 2018), and TweetEval-Sentiment (Barbieri et al. 2020). For pair classification, we

Algorithm 1 HeatGeo Training

Require: Corpus $\mathcal{D}$, teacher T , student S_θ , graph size k , diffusion scales $\mathcal{R}$

- 1: Cache teacher embeddings $t_i = T(x_i)$ for all $x_i \in \mathcal{D}$
- 2: Build a mutual k NN teacher graph W^T
- 3: Compute transition matrix $P^T = (D^T)^{-1} W^T$
- 4: Precompute sparse diffusion candidates and targets $p_{i,r}^T$
- 5: Compute spectral coordinates u_i^T from the graph Laplacian
- 6: **for** each training epoch **do**
- 7: Activate diffusion scales according to the curriculum
- 8: **for** each mini-batch B **do**
- 9: Encode anchors and candidate texts with S_θ
- 10: Compute student candidate distributions p_i^S
- 11: Compute $\mathcal{L}_{\text{diff}}$, $\mathcal{L}_{\text{spec}}$, and $\mathcal{L}_{\text{anchor}}$
- 12: Update θ by minimizing $\mathcal{L}_{\text{HeatGeo}}$
- 13: **end for**
- 14: **end for**
- 15: **return** Distilled student encoder S_θ

report average precision (AP) on MRPC (Dolan and Brockett 2005), SciTail (Khot, Sabharwal, and Clark 2018), and WiC (Pilehvar and Camacho-Collados 2019). For semantic textual similarity, we report Spearman correlation on SICK (Marelli et al. 2014), STS12 (Agirre et al. 2012), and STS-Benchmark (Cer et al. 2017). Emotion, WiC, and STS-Benchmark are treated as in-domain benchmarks; the remaining datasets are treated as out-of-domain transfer benchmarks.

Models. We keep two teacher-student configurations: BGE-M3 (Chen et al. 2024) $\rightarrow$ BERT-base (Devlin et al. 2019) and Qwen3-Embedding-4B (Zhang et al. 2025) $\rightarrow$ BERT-base. Teacher embeddings are cached once before HeatGeo training. After graph construction, the teacher model is removed from GPU memory; student training uses cached teacher embeddings, sparse diffusion targets, and spectral coordinates.

Training data. The distillation corpus is constructed from unlabeled text drawn from the in-domain training split following the TALAS protocol. For pairwise datasets, both sides of each pair are flattened into the text corpus before graph construction. HeatGeo builds the teacher semantic graph over this corpus and trains without using downstream labels.

Baselines. We compare HeatGeo with the full teacher, the undistilled BERT-base student, SimCSE-unsup (Gao, Yao, and Chen 2021), cross-tokenizer distillation methods CDM (Chen et al. 2025) and DSKD (Zhang et al. 2024b), sentence embedding distillation methods Jasper and Stella (Zhang et al. 2024a) and DistillCSE (Gao et al. 2021; Xu et al. 2023), EMO (Truong et al. 2025), and TALAS (Dao et al. 2026). Component-level comparisons against pointwise anchoring and mini-batch relational matching are included in the ablation study.

Evaluation protocol. For classification, the encoder is frozen and a logistic regression classifier is trained on top of the sentence embeddings. For pair classification and STS, examples are embedded independently and compared by cosine similarity. We report per-task scores, the average over in-domain tasks, the average over out-of-domain tasks, and the overall average across all benchmarks.

Implementation details. Unless otherwise specified, HeatGeo uses graph size $k = 50$, graph temperature $\tau_g = 0.1$, diffusion scales $\{1, 2, 4\}$, candidate size 64, 32 diffusion positives, 16 hard negatives, 16 random negatives, spectral dimension 16, and five training epochs. The scale weights are $(1.0, 0.5, 0.25)$. The loss weights are $\lambda_1 = 1.0$, $\lambda_2 = 0.1$, and $\lambda_3 = 0.05$.

4.2 Main Results

Tables 1 and 2 define the main reporting format for the two teacher-student pairs. The tables follow TALAS by grouping tasks into classification, pair classification, and STS, while separately reporting in-domain and out-of-domain averages. This format tests whether HeatGeo preserves embedding quality on both the training-related domain and held-out transfer domains.

4.3 Ablation Study

Table 3 isolates the contribution of each HeatGeo component. The ablations separate the effect of multi-scale diffusion, spectral manifold anchoring, the pointwise anchor, curriculum scheduling, and candidate composition. This distinguishes corpus-level diffusion transfer from simpler local alignment objectives.

Table 3: Ablation template for HeatGeo. Checkmarks indicate enabled components. Avg-In, Avg-Out, and Avg-All follow the domain split in Tables 1 and 2.

Variant	$\mathcal{L}_{\text{diff}}$	$\mathcal{L}_{\text{spec}}$	$\mathcal{L}_{\text{anchor}}$	Curr.	Full C_i	Avg-In	Avg-Out	Avg-All
HeatGeo full	✓	✓	✓	✓	✓	–	–	–
w/o diffusion loss	–	✓	✓	✓	✓	–	–	–
w/o spectral anchoring	✓	–	✓	✓	✓	–	–	–
w/o anchor	✓	✓	–	✓	✓	–	–	–
w/o curriculum	✓	✓	✓	–	✓	–	–	–
w/o hard negatives	✓	✓	✓	✓	partial	–	–	–
w/o random negatives	✓	✓	✓	✓	partial	–	–	–
single scale $r = 1$	✓	✓	✓	–	✓	–	–	–
single scale $r = 4$	✓	✓	✓	–	✓	–	–	–
pointwise anchor only	–	–	✓	–	–	–	–	–
mini-batch relational	–	–	✓	–	–	–	–	–

Table 1: Main evaluation template for BGE-M3 $\rightarrow$ BERT-base. Classification tasks report macro-F1, pair classification tasks report average precision (AP), and STS tasks report Spearman correlation. Starred datasets are in-domain.

	Classification (F1)			Pair Classification (AP)			STS (Spearman)			Domain Avg		Avg-All
	Banking77	TweetEval	Emotion*	MRPC	SciTail	WiC*	SICK	STS12	STSB*	Avg-In	Avg-Out	Avg-All
BGE-M3 $\rightarrow$ BERT-base												
Teacher	–	–	–	–	–	–	–	–	–	–	–	–
Student base	–	–	–	–	–	–	–	–	–	–	–	–
SimCSE-unsup	–	–	–	–	–	–	–	–	–	–	–	–
CDM	–	–	–	–	–	–	–	–	–	–	–	–
DSKD	–	–	–	–	–	–	–	–	–	–	–	–
Jasper/Stella	–	–	–	–	–	–	–	–	–	–	–	–
DistilCSE	–	–	–	–	–	–	–	–	–	–	–	–
EMO	–	–	–	–	–	–	–	–	–	–	–	–
TALAS	–	–	–	–	–	–	–	–	–	–	–	–
HeatGeo	–	–	–	–	–	–	–	–	–	–	–	–

Table 2: Main evaluation results for Qwen3-Embedding-4B $\rightarrow$ BERT-base. Classification tasks report macro-F1, pair classification tasks report average precision (AP), and STS tasks report Spearman correlation. Starred datasets are in-domain.

	Classification (F1)			Pair Classification (AP)			STS (Spearman)			Domain Avg		Avg-All
	Banking77	TweetEval	Emotion*	MRPC	SciTail	WiC*	SICK	STS12	STSB*	Avg-In	Avg-Out	Avg-All
Qwen3-Embedding-4B $\rightarrow$ BERT-base												
Teacher	93.63	72.22	68.94	84.79	91.77	66.80	83.43	82.55	86.87	74.20	84.73	81.22
Student base	84.86	47.79	68.02	77.36	67.74	59.22	42.43	21.54	20.30	42.44	60.33	54.36
SimCSE-unsup	89.21	69.13	53.55	82.86	76.59	71.64	68.19	61.68	70.36	65.18	74.61	71.47
CDM	89.75	70.58	53.11	84.83	75.19	71.06	69.37	68.48	73.92	66.03	76.37	72.92
DSKD	89.66	69.93	54.51	83.13	77.95	71.53	68.65	63.37	71.42	65.82	75.45	72.24
Jasper/Stella	89.71	73.83	65.98	85.33	80.99	70.50	73.96	67.90	75.65	70.71	78.62	75.98
DistilCSE	90.94	70.90	60.80	82.86	78.98	68.63	75.12	72.61	77.08	68.84	78.57	75.32
EMO	90.78	73.54	67.48	84.46	81.57	69.76	75.66	68.83	77.17	71.47	79.14	76.58
TALAS	92.34	73.74	64.41	85.97	82.66	68.65	78.38	72.27	79.98	71.01	80.89	77.60
HeatGeo	92.62	72.27	66.42	83.57	89.15	66.43	79.69	70.43	81.79	71.55	81.29	78.04

5 Discussion

HeatGeo is designed around the observation that embedding distillation should match the object that retrieval systems actually use: corpus-level neighborhood geometry. The method has three practical advantages. First, it uses cached teacher embeddings rather than teacher hidden states, attention maps, or online teacher forward passes, which lowers training-time memory pressure. Second, it is tokenizer-agnostic because the loss is defined over final embeddings and graph diffusion targets. Third, it does not change the student architecture, so the distilled model can be deployed as an ordinary text encoder.

The main limitation is that HeatGeo requires a corpus-level preprocessing step. Building a teacher graph is worthwhile when the same training corpus supports substantial distillation, but it adds overhead compared with purely mini-batch objectives. HeatGeo also depends on the quality and coverage of the teacher graph: if the teacher embeddings

contain spurious neighborhoods, diffusion may propagate those errors. Future work can study approximate nearest-neighbor construction, adaptive diffusion scales, and online graph refresh for changing corpora.

6 Conclusion

We introduced HeatGeo, a corpus-level heat-kernel geometry distillation framework for compact text embedding models. Instead of matching only teacher vectors, token states, or mini-batch similarity snapshots, HeatGeo distills multi-scale semantic diffusion over a teacher-induced mutual k NN graph and anchors global topology with spectral coordinates. This directly targets the neighborhood geometry that embedding models use in retrieval, clustering, and transfer settings while requiring only cached teacher embeddings and no student architecture changes. Together, the diffusion, spectral, and anchor objectives position HeatGeo as a practical and retrieval-aligned alternative to local embedding dis-

tillation objectives.

References

- Agarwal, R.; Vieillard, N.; Zhou, Y.; Stanczyk, P.; Ramos Garea, S.; Geist, M.; and Bachem, O. 2024. On-Policy Distillation of Language Models: Learning from Self-Generated Mistakes. In *The Twelfth International Conference on Learning Representations*.
- Agirre, E.; Cer, D.; Diab, M.; and Gonzalez-Agirre, A. 2012. SemEval-2012 Task 6: A Pilot on Semantic Textual Similarity. In *Proceedings of the First Joint Conference on Lexical and Computational Semantics*, 385–393.
- Akram, M. K.; Sturua, S.; Havriushenko, N.; Herreros, Q.; Günther, M.; Werk, M.; and Xiao, H. 2026. jina-embeddings-v5-text: Task-Targeted Embedding Distillation. *arXiv preprint arXiv:2602.15547*.
- Alqahtani, S.; Lalwani, G.; Zhang, Y.; Romeo, S.; and Mansour, S. 2021. Using Optimal Transport as Alignment Objective for Fine-tuning Multilingual Contextualized Embeddings. *arXiv preprint arXiv:2110.02887*.
- Anshumann; Zaidi, M. A.; Kedia, A.; Ahn, J.; Kwon, T.; Lee, K.; Lee, H.; and Lee, J. 2025. Sparse Logit Sampling: Accelerating Knowledge Distillation in LLMs. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics*.
- Barbieri, F.; Camacho-Collados, J.; Espinosa-Anke, L.; and Neves, L. 2020. TweetEval: Unified Benchmark and Comparative Evaluation for Tweet Classification. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, 1644–1650.
- Belkin, M.; and Niyogi, P. 2003. Laplacian Eigenmaps for Dimensionality Reduction and Data Representation. *Neural Computation*, 15(6): 1373–1396.
- Casanueva, I.; Temcinas, T.; Gerz, D.; Henderson, M.; and Vulic, I. 2020. Efficient Intent Detection with Dual Sentence Encoders. In *Proceedings of the 2nd Workshop on Natural Language Processing for Conversational AI*, 38–45.
- Cer, D.; Diab, M.; Agirre, E.; Lopez-Gazpio, I.; and Specia, L. 2017. SemEval-2017 Task 1: Semantic Textual Similarity Multilingual and Crosslingual Focused Evaluation. In *Proceedings of the 11th International Workshop on Semantic Evaluation*, 1–14.
- Chen, J.; Xiao, S.; Zhang, P.; Luo, K.; Lian, D.; and Liu, Z. 2024. BGE M3-Embedding: Multi-Lingual, Multi-Functionality, Multi-Granularity Text Embeddings Through Self-Knowledge Distillation. *arXiv preprint arXiv:2402.03216*.
- Chen, Y.; Liu, Y.; Meng, F.; Chen, Y.; Xu, J.; and Zhou, J. 2025. Enhancing Cross-Tokenizer Knowledge Distillation with Contextual Dynamical Mapping. *arXiv preprint arXiv:2502.11104*.
- Chung, I.; Kerboua, I.; Kardos, M.; Solomatin, R.; and Enevoldsen, K. 2025. Maintaining MTEB: Towards Long Term Usability and Reproducibility of Embedding Benchmarks. *arXiv preprint arXiv:2506.21182*.
- Coifman, R. R.; and Lafon, S. 2006. Diffusion Maps. *Applied and Computational Harmonic Analysis*, 21(1): 5–30.
- Dao, Q. P.; Nguyen, H. S.; Pham, K. C.; Ngo Van, L.; Nguyen, D. T.-N.; Nguyen, T. H.; and Le, T. 2026. TALAS: Teacher-Anchored Layer Alignment with Adaptive Sharpness-Aware Minimization for Embedding Distillation. Manuscript.
- Devlin, J.; Chang, M.-W.; Lee, K.; and Toutanova, K. 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics*, 4171–4186.
- Dolan, W. B.; and Brockett, C. 2005. Automatically Constructing a Corpus of Sentential Paraphrases. In *Proceedings of the Third International Workshop on Paraphrasing*.
- Gao, C.; Wu, X.; Wang, P.; Wang, J.; Zang, L.; Wang, Z.; and Hu, S. 2021. DistilCSE: Effective Knowledge Distillation for Contrastive Sentence Embeddings. *arXiv preprint arXiv:2112.05638*.
- Gao, T.; Yao, X.; and Chen, D. 2021. SimCSE: Simple Contrastive Learning of Sentence Embeddings. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, 6894–6910.
- Hinton, G.; Vinyals, O.; and Dean, J. 2015. Distilling the Knowledge in a Neural Network. *arXiv preprint arXiv:1503.02531*.
- Jiao, X.; Yin, Y.; Shang, L.; Jiang, X.; Chen, X.; Li, L.; Wang, F.; and Liu, Q. 2020. TinyBERT: Distilling BERT for Natural Language Understanding. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, 4163–4174.
- Khot, T.; Sabharwal, A.; and Clark, P. 2018. SciTail: A Textual Entailment Dataset from Science Question Answering. In *Proceedings of the AAAI Conference on Artificial Intelligence*.
- Kim, Y.; Shin, D.; Kang, M.; Na, B.; and Moon, I.-C. 2025. Distillation of Large Language Models via Concrete Score Matching. *arXiv preprint arXiv:2509.25837*.
- Ko, J.; Chen, T.; Kim, S.; Ding, T.; Liang, L.; Zharkov, I.; and Yun, S.-Y. 2025. DistiLLM-2: A Contrastive Approach Boosts the Distillation of LLMs. In *Proceedings of the 42nd International Conference on Machine Learning*.
- Kornblith, S.; Norouzi, M.; Lee, H.; and Hinton, G. 2019. Similarity of Neural Network Representations Revisited. In *Proceedings of the 36th International Conference on Machine Learning*, 3519–3529.
- Lee, J.; Dai, Z.; Ren, X.; Chen, B.; Cer, D.; Cole, J. R.; Hui, K.; Boratko, M.; Kapadia, R.; Ding, W.; Luan, Y.; Duddu, S. M. K.; Abrego, G. H.; Shi, W.; Gupta, N.; Kusupati, A.; Jain, P.; Jonnalagadda, S. R.; Chang, M.-W.; and Naim, I. 2024. Gecko: Versatile Text Embeddings Distilled from Large Language Models. *arXiv preprint arXiv:2403.20327*.
- Marelli, M.; Menini, S.; Baroni, M.; Bentivogli, L.; Bernardi, R.; and Zamparelli, R. 2014. A SICK cure for the evaluation of compositional distributional semantic models. In *Proceedings of the 9th International Workshop on Semantic Evaluation*, 216–223.

Muennighoff, N.; Tazi, N.; Magne, L.; and Reimers, N. 2023. MTEB: Massive Text Embedding Benchmark. In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*, 2014–2037.

Park, W.; Kim, D.; Lu, Y.; and Cho, M. 2019. Relational Knowledge Distillation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 3967–3976.

Pilehvar, M. T.; and Camacho-Collados, J. 2019. WiC: The Word-in-Context Dataset for Evaluating Context-Sensitive Meaning Representations. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics*, 1267–1273.

Reimers, N.; and Gurevych, I. 2019. Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, 3982–3992.

Sanh, V.; Debut, L.; Chaumond, J.; and Wolf, T. 2019. DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter. *arXiv preprint arXiv:1910.01108*.

Saravia, E.; Liu, H.-C. T.; Huang, Y.-H.; Wu, J.; and Chen, Y.-S. 2018. CARER: Contextualized Affect Representations for Emotion Recognition. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing Workshop on Computational Approaches to Subjectivity, Sentiment and Social Media Analysis*, 368–379.

Thakur, N.; Reimers, N.; Rücklé, A.; Srivastava, A.; and Gurevych, I. 2021. BEIR: A Heterogeneous Benchmark for Zero-shot Evaluation of Information Retrieval Models. In *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*.

Tian, Y.; Pei, S.; Zhang, X.; Zhang, C.; and Chawla, N. V. 2025. Knowledge Distillation on Graphs: A Survey. *ACM Computing Surveys*, 57(8): 189:1–189:16.

Truong, M.-P.; Vu, H. A.; Vu, T.; Diep, N. T. N.; Van, L. N.; Nguyen, T. H.; and Le, T. 2025. EMO: Embedding Model Distillation via Intra-Model Relation and Optimal Transport Alignments. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, 7594–7606.

Vujanic, R.; and Ruecksties, T. 2025. LEAF: Knowledge Distillation of Text Embedding Models with Teacher-Aligned Representations. *arXiv preprint arXiv:2509.12539*.

Wang, W.; Wei, F.; Dong, L.; Bao, H.; Yang, N.; and Zhou, M. 2020. MiniLM: Deep Self-Attention Distillation for Task-Agnostic Compression of Pre-Trained Transformers. In *Advances in Neural Information Processing Systems*.

Xu, J.; Shao, W.; Chen, L.; and Liu, L. 2023. DistillCSE: Distilled Contrastive Learning for Sentence Embeddings. *arXiv preprint arXiv:2310.13499*.

Xu, X.; Li, M.; Tao, C.; Shen, T.; Cheng, R.; Li, J.; Xu, C.; Tao, D.; and Zhou, T. 2024. A Survey on Knowledge Distillation of Large Language Models. *arXiv preprint arXiv:2402.13116*.

Zhang, D.; Li, J.; Zeng, Z.; and Wang, F. 2024a. Jasper and Stella: Distillation of SOTA Embedding Models. *arXiv preprint arXiv:2412.19048*.

Zhang, D.; Zou, P.; and Zhou, Y. 2025. Dewey Long Context Embedding Model: A Technical Report. *arXiv preprint arXiv:2503.20376*.

Zhang, S.; Liu, Y.; Sun, Y.; and Shah, N. 2022. Graph-less Neural Networks: Teaching Old MLPs New Tricks via Distillation. In *The Tenth International Conference on Learning Representations*.

Zhang, S.; Zhang, X.; Sun, Z.; Chen, Y.; and Xu, J. 2024b. Dual-Space Knowledge Distillation for Large Language Models. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, 18164–18181.

Zhang, Y.; Li, M.; Long, D.; Zhang, X.; Lin, H.; Yang, B.; Xie, P.; Yang, A.; Liu, D.; Lin, J.; Huang, F.; and Zhou, J. 2025. Qwen3 Embedding: Advancing Text Embedding and Reranking Through Foundation Models. *arXiv preprint arXiv:2506.05176*.

A Theoretical Proofs

This appendix provides the proofs for the theoretical statements in Section 3. For a distribution p on a finite set E and a subset $A \subseteq E$, write $p(A) = \sum_{j \in A} p(j)$, and define total variation distance by

$$\text{TV}_E(p, q) = \sup_{A \subseteq E} |p(A) - q(A)| = \frac{1}{2} \sum_{j \in E} |p(j) - q(j)|. \quad (29)$$

A.1 Proof of Lemma 1

For every subset $A \subseteq C_i$,

$$|p_i^S(A) - p_{i,r}^T(A)| \leq \text{TV}_{C_i}(p_i^S, p_{i,r}^T) \quad (30)$$

by the variational definition of total variation distance on C_i . Since $p_{i,r}^T$ and p_i^S are both distributions over C_i , Pinsker’s inequality gives

$$\begin{aligned} \text{TV}_{C_i}(p_i^S, p_{i,r}^T) &\leq \sqrt{\frac{1}{2} D_{\text{KL}}(p_{i,r}^T \| p_i^S)} \\ &\leq \sqrt{\frac{\delta}{2}}. \end{aligned} \quad (31)$$

Combining the two inequalities proves the claim.

A.2 Proof of Lemma 2

On C_i , the full diffusion row decomposes as

$$q_{i,r}(j) = \rho p_{i,r}^T(j), \quad j \in C_i, \quad (32)$$

where $\rho = \rho_{i,r}(C_i)$. Outside C_i , the zero-extended student distribution has no mass. Therefore

$$\begin{aligned} 2 \text{TV}_{\mathcal{D}}(q_{i,r}, \hat{p}_i^S) &= \sum_{j \in C_i} |\rho p_{i,r}^T(j) - p_i^S(j)| + \sum_{j \notin C_i} q_{i,r}(j). \end{aligned} \quad (33)$$

For $j \in C_i$,

$$\begin{aligned} |\rho p_{i,r}^T(j) - p_i^S(j)| &\leq \rho |p_{i,r}^T(j) - p_i^S(j)| \\ &\quad + (1 - \rho) p_i^S(j). \end{aligned} \quad (34)$$

Summing over $j \in C_i$ gives

$$\begin{aligned} \sum_{j \in C_i} |\rho p_{i,r}^T(j) - p_i^S(j)| \\ \leq 2\rho \text{TV}_{C_i}(p_{i,r}^T, p_i^S) + (1 - \rho). \end{aligned} \quad (35)$$

Since $\sum_{j \notin C_i} q_{i,r}(j) = 1 - \rho$, we obtain

$$\begin{aligned} 2 \text{TV}_{\mathcal{D}}(q_{i,r}, \hat{p}_i^S) \\ \leq 2\rho \text{TV}_{C_i}(p_{i,r}^T, p_i^S) + 2(1 - \rho). \end{aligned} \quad (36)$$

Dividing by two proves

$$\text{TV}_{\mathcal{D}}(q_{i,r}, \hat{p}_i^S) \leq (1 - \rho) + \rho \text{TV}_{C_i}(p_{i,r}^T, p_i^S). \quad (37)$$

If $\rho \geq 1 - \epsilon$, then $1 - \rho \leq \epsilon$ and $\rho \leq 1$. Applying Lemma 1 to the restricted mismatch yields

$$\text{TV}_{\mathcal{D}}(q_{i,r}, \hat{p}_i^S) \leq \epsilon + \sqrt{\frac{\delta}{2}}. \quad (38)$$

A.3 Proof of Proposition 3

The mean-square preservation statement follows directly from the definition of $\mathcal{L}_{\text{spec}}(B)$. For any $i, j \in B$, the reverse triangle inequality gives

$$\begin{aligned} \left| \|z_i^S - z_j^S\|_2 - \|u_i^T - u_j^T\|_2 \right| \\ \leq \|(z_i^S - z_j^S) - (u_i^T - u_j^T)\|_2. \end{aligned} \quad (39)$$

The triangle inequality then gives

$$\begin{aligned} \|(z_i^S - z_j^S) - (u_i^T - u_j^T)\|_2 \\ \leq \|z_i^S - u_i^T\|_2 + \|z_j^S - u_j^T\|_2. \end{aligned} \quad (40)$$

Averaging over all ordered pairs and applying Jensen's inequality yields

$$\begin{aligned} \frac{1}{|B|^2} \sum_{i,j \in B} (\|z_i^S - u_i^T\|_2 + \|z_j^S - u_j^T\|_2) \\ = \frac{2}{|B|} \sum_{i \in B} \|z_i^S - u_i^T\|_2 \leq 2\sqrt{\eta}. \end{aligned} \quad (41)$$

This proves the proposition.

A.4 Proof of Corollary 4

Let

$$\begin{aligned} \mathcal{L}_{\text{HeatGeo}}(B) &= \lambda_1 \mathcal{L}_{\text{diff}}(B) + \lambda_2 \mathcal{L}_{\text{spec}}(B) \\ &\quad + \lambda_3 \mathcal{L}_{\text{anchor}}(B) \leq \gamma. \end{aligned} \quad (42)$$

All three losses are nonnegative. Hence

$$\begin{aligned} \mathcal{L}_{\text{diff}}(B) &\leq \frac{\gamma}{\lambda_1}, \\ \mathcal{L}_{\text{spec}}(B) &\leq \frac{\gamma}{\lambda_2}, \\ \mathcal{L}_{\text{anchor}}(B) &\leq \frac{\gamma}{\lambda_3}. \end{aligned} \quad (43)$$

Let $\Omega_e = \sum_{r \in \mathcal{R}_e} \omega_r$. Draw I uniformly from B and draw R from $\mathcal{R}_e$ with probability $\Pr(R = r) = \omega_r / \Omega_e$. By the definition of $\mathcal{L}_{\text{diff}}$,

$$\begin{aligned} \mathbb{E}_{I,R} [D_{\text{KL}}(p_{I,R}^T \| p_I^S)] \\ = \frac{\mathcal{L}_{\text{diff}}(B)}{\Omega_e} \leq \frac{\gamma}{\lambda_1 \Omega_e}. \end{aligned} \quad (44)$$

Applying Lemma 1 and Jensen's inequality gives

$$\begin{aligned} \mathbb{E}_{I,R} \left[\sup_{A \subseteq C_I} |p_I^S(A) - p_{I,R}^T(A)| \right] \\ \leq \sqrt{\frac{\gamma}{2\lambda_1 \Omega_e}}. \end{aligned} \quad (45)$$

Combining this bound with Lemma 2 and the assumption $\rho_{i,r}(C_i) \geq 1 - \epsilon$ gives

$$\begin{aligned} \mathbb{E}_{I,R} [\text{TV}_{\mathcal{D}}(q_{I,R}, \hat{p}_I^S)] \\ \leq \epsilon + \sqrt{\frac{\gamma}{2\lambda_1 \Omega_e}}. \end{aligned} \quad (46)$$

Proposition 3 with $\eta = \gamma / \lambda_2$ gives

$$\begin{aligned} \frac{1}{|B|^2} \sum_{i,j \in B} \left| \|z_i^S - z_j^S\|_2 - \|u_i^T - u_j^T\|_2 \right| \\ \leq 2\sqrt{\frac{\gamma}{\lambda_2}}, \end{aligned} \quad (47)$$

and the anchor bound follows directly:

$$\frac{1}{|B|} \sum_{i \in B} (1 - \cos(W_a s_i, t_i)) \leq \frac{\gamma}{\lambda_3}. \quad (48)$$